Diabetes Analysis with Bayesian Network

Fundamentals Of AI - Module 3
Professor Paolo Torroni

Sara Tajernia
Mohammad Choupan
Mohammad Amin Mirzakuchaki

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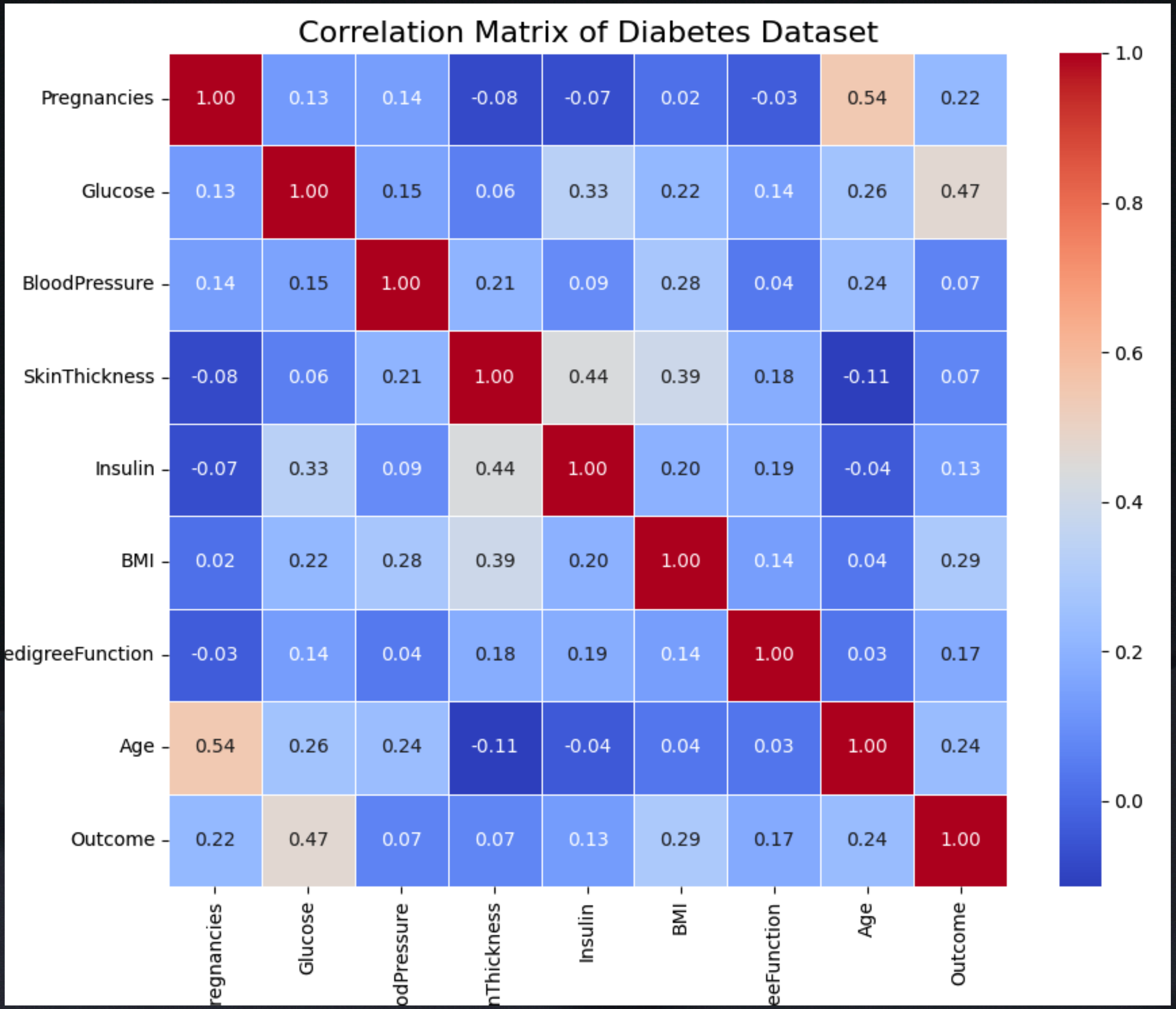
Dataset Overview

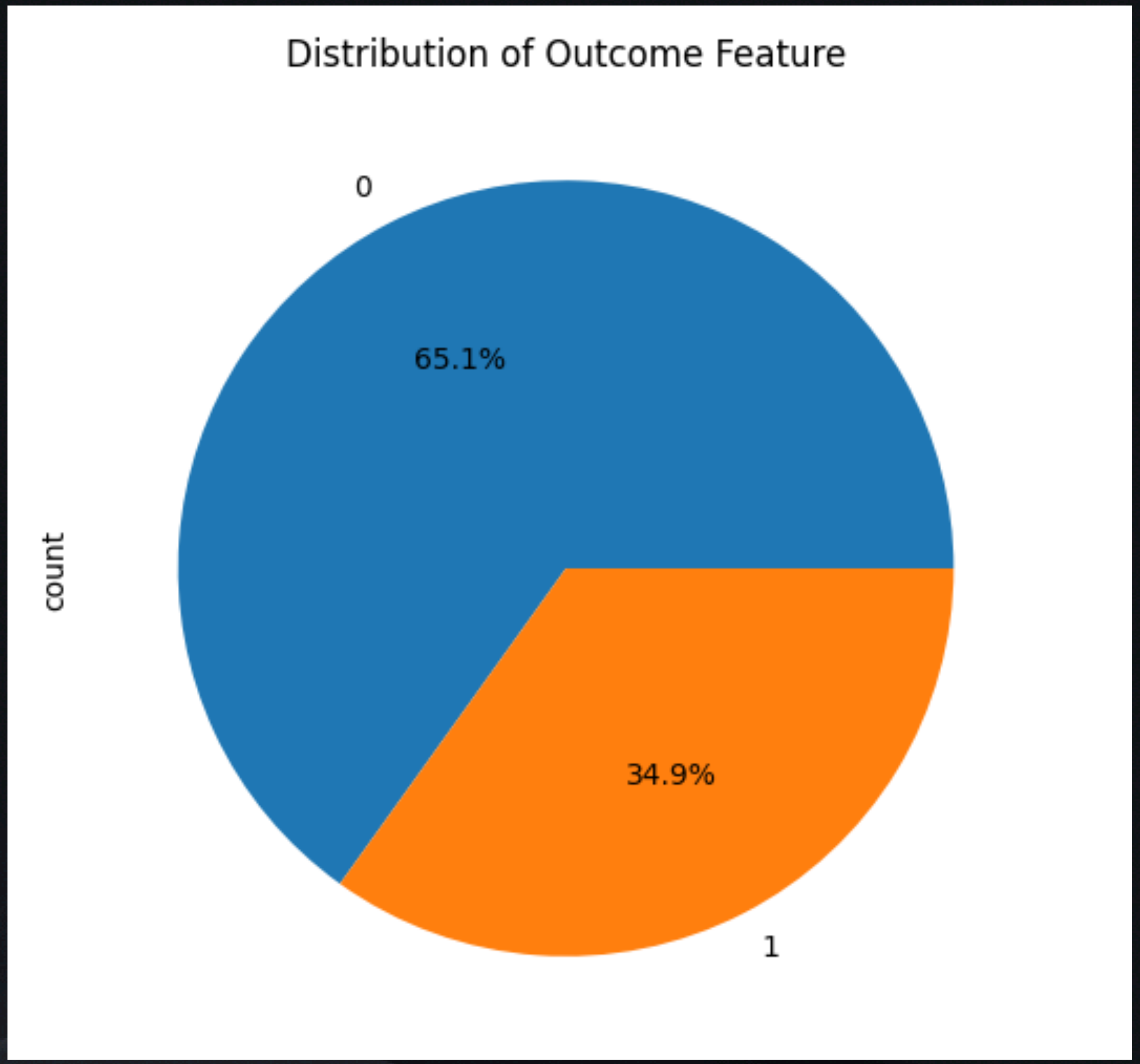
Attribute	*Description*
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration 2 hours after an oral glucose tolerance test
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-hour serum insulin (μU/ml)
BMI	Body mass index (weight in kg / height in m^2)
DiabetesPedigreeFunction	Diabetes pedigree function
Age	Age (years)
Outcome	Class variable (0 or 1) indicating if the patient has diabetes (1) or not (0)

Exploratory Data Analysis

- statistical summary of the dataset
- plotting distributions of features
- checking for missing values
- checking for zero values in features where it is not possible
- scatter plot of each feature vs Outcome
- pair plot of features
- heatmap of correlation matrix

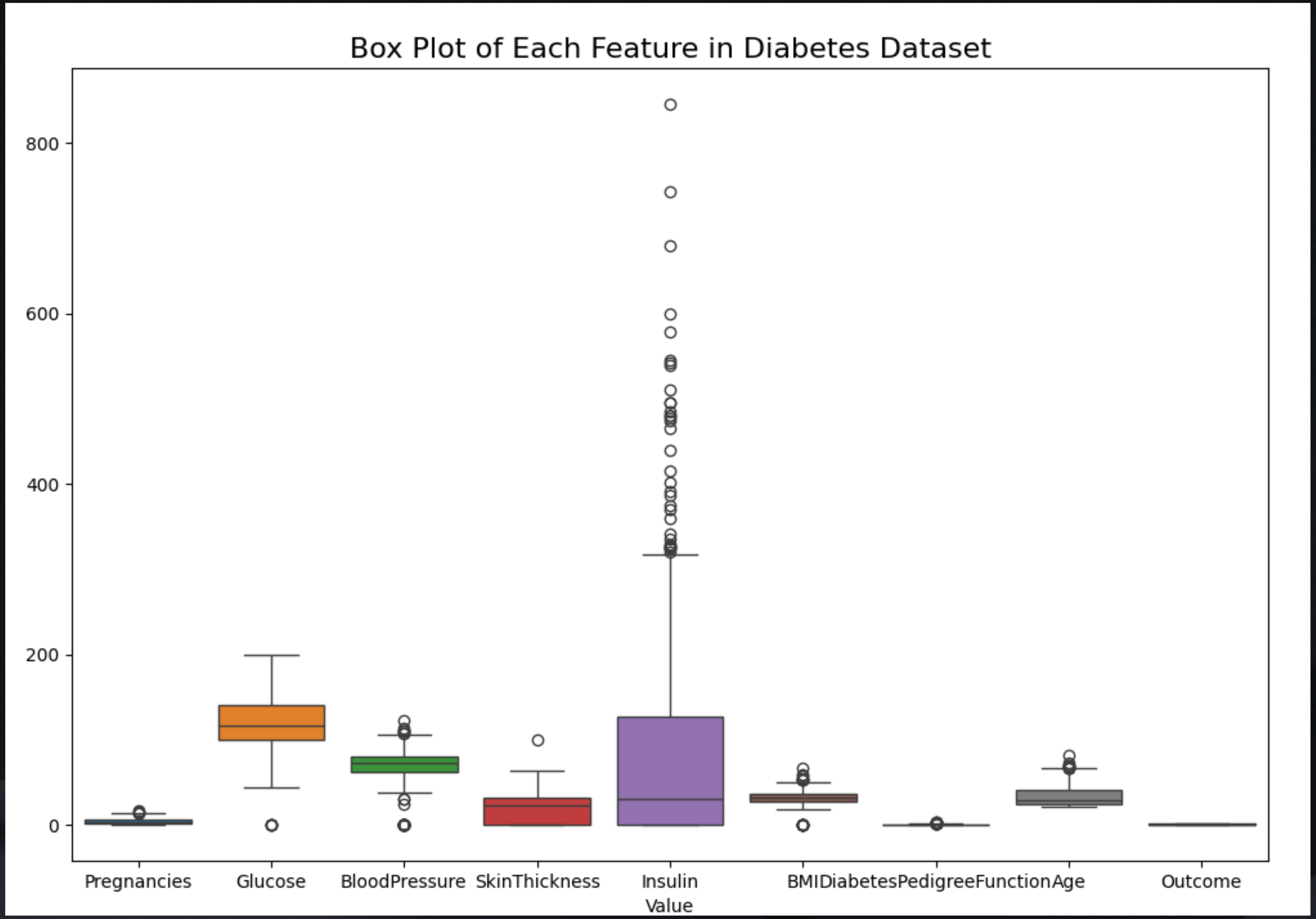
Data Sourcing

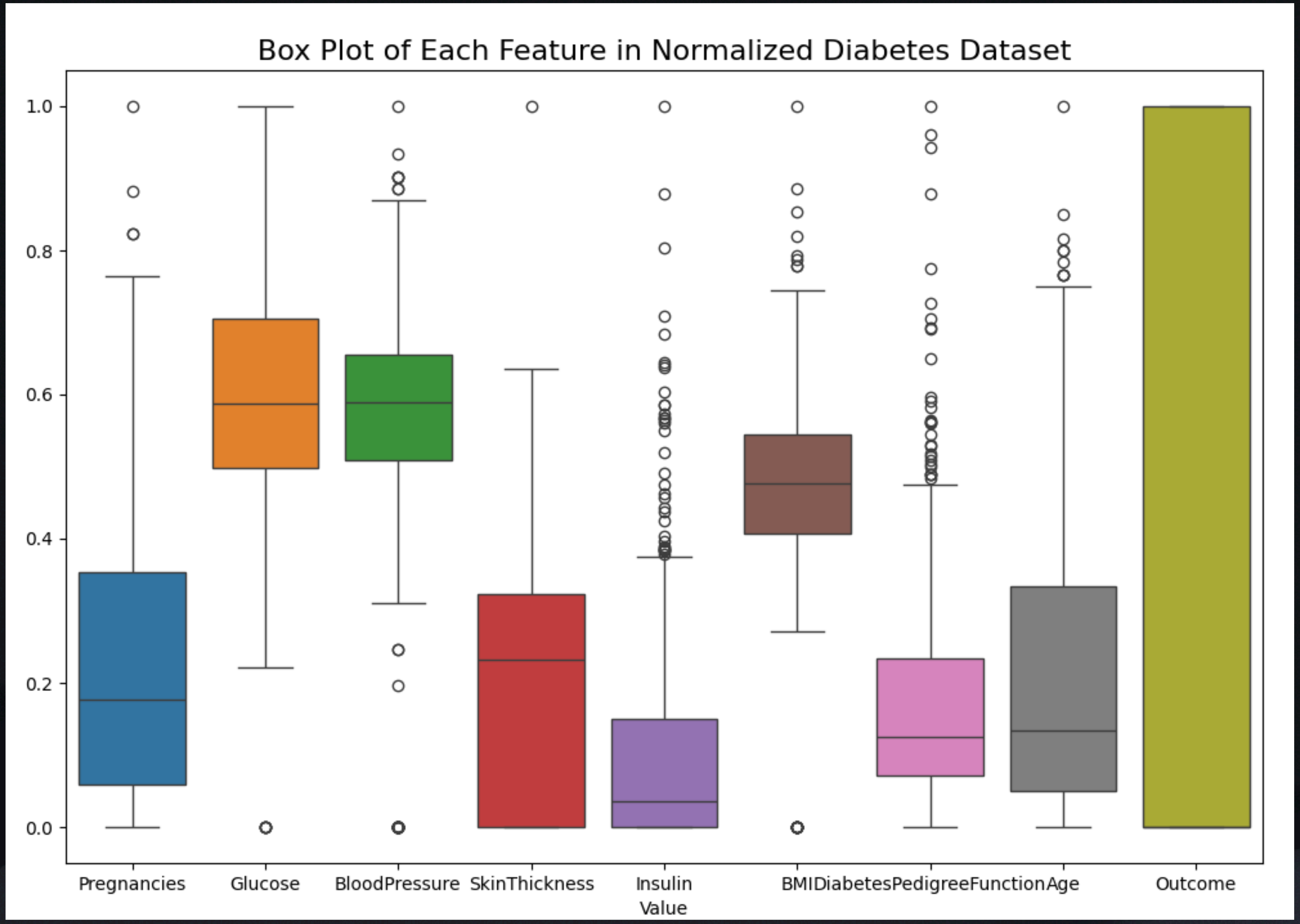




- 768 total instances
- 500 non-diabetic (0), 268 diabetic (1)
- Mix of continuous and categorical features

Normalize (MinMax Scaler)



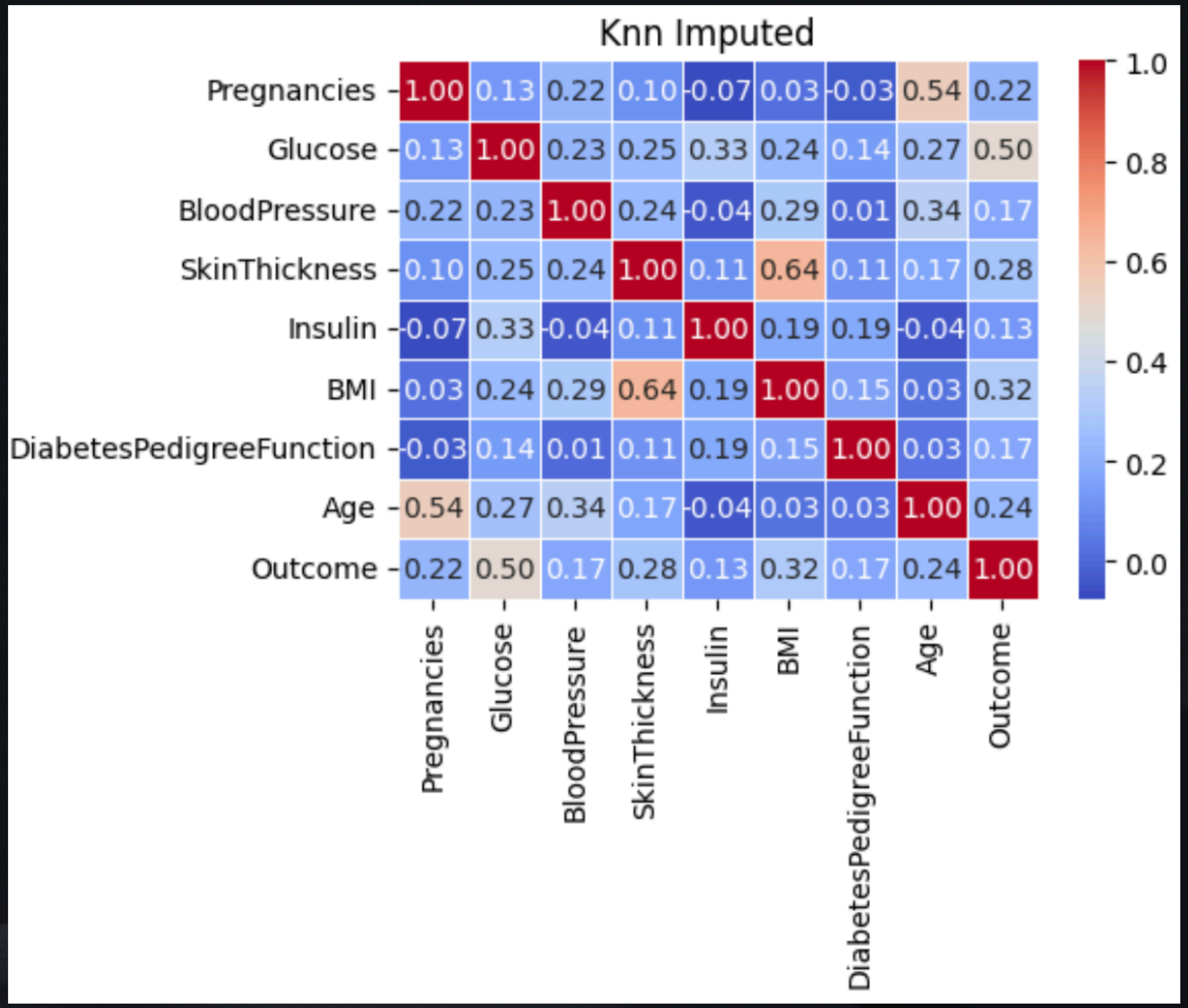


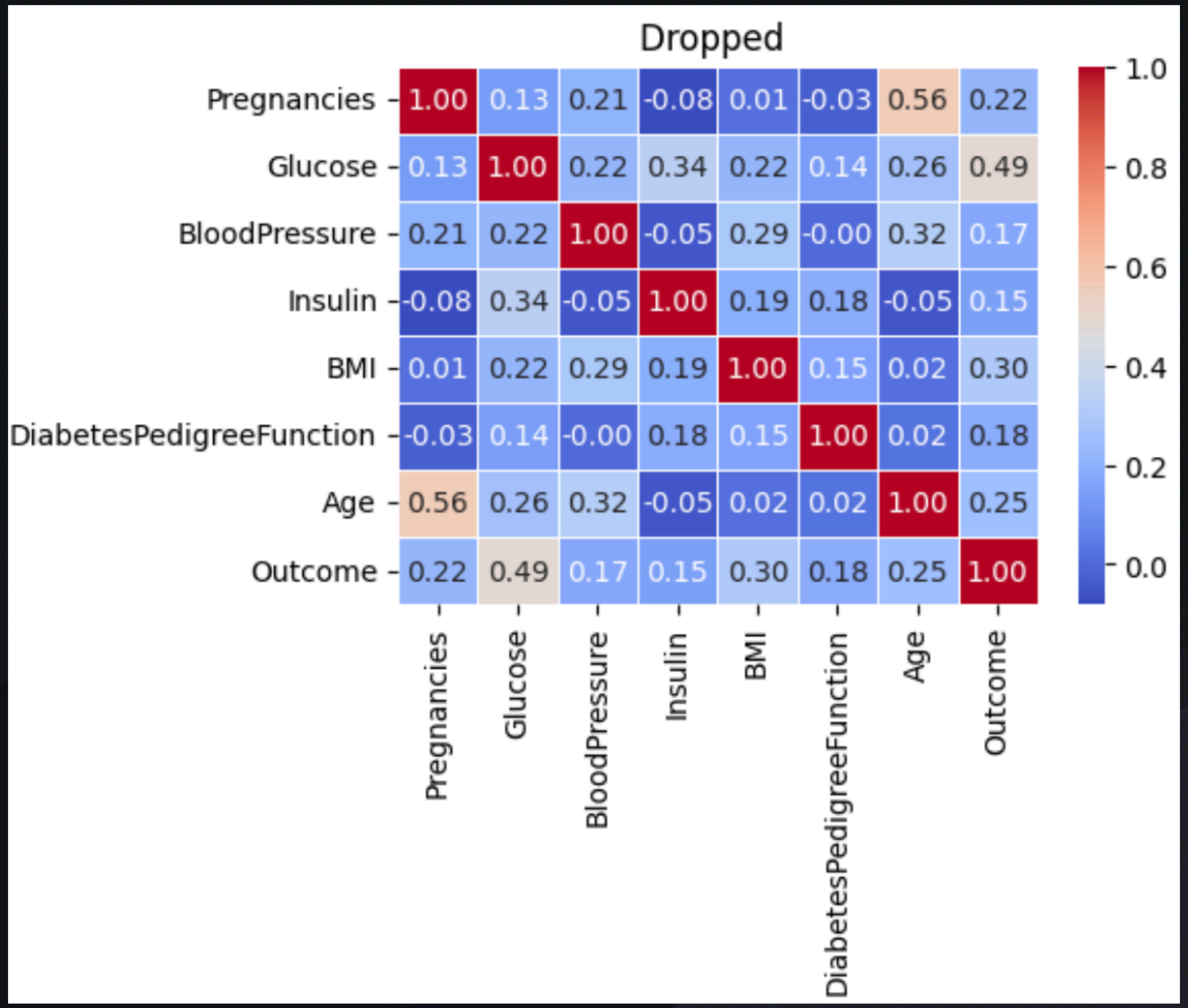
Preprocessing

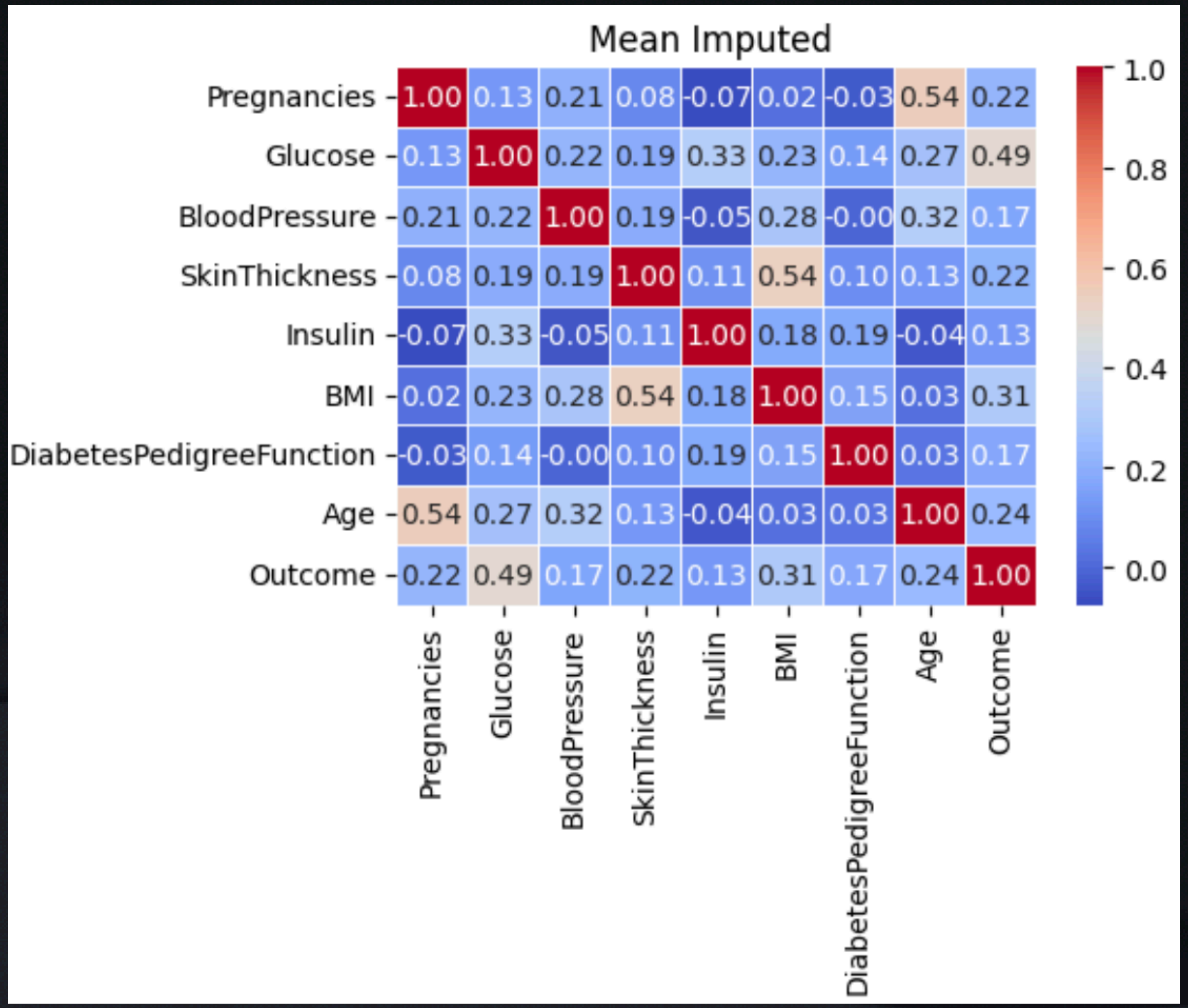
- *Handel Zero Values*
  - Remove rows with missing values ✓
  - **Remove column with most missing values and remove rows with missing values in other columns ✓ (SkinThickness)**
  - *Impute missing values with mean/median/mode*
  - *Use advanced imputation techniques like KNN imputation or regression imputation*

{'Glucose': 5, 'BloodPressure': 35, 'SkinThickness': 227, 'BMI': 11, 'Age': 0}

Handle Missing Values



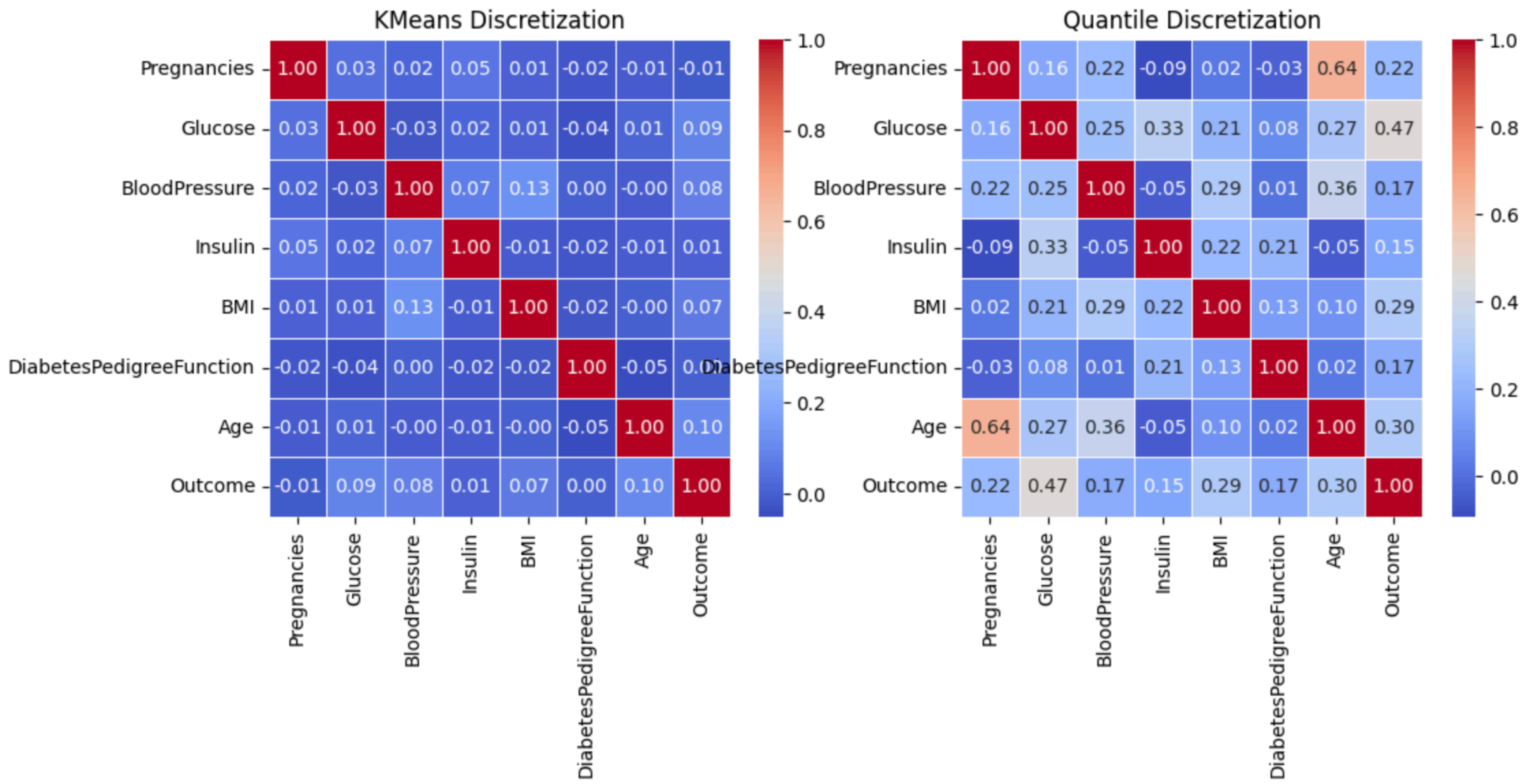




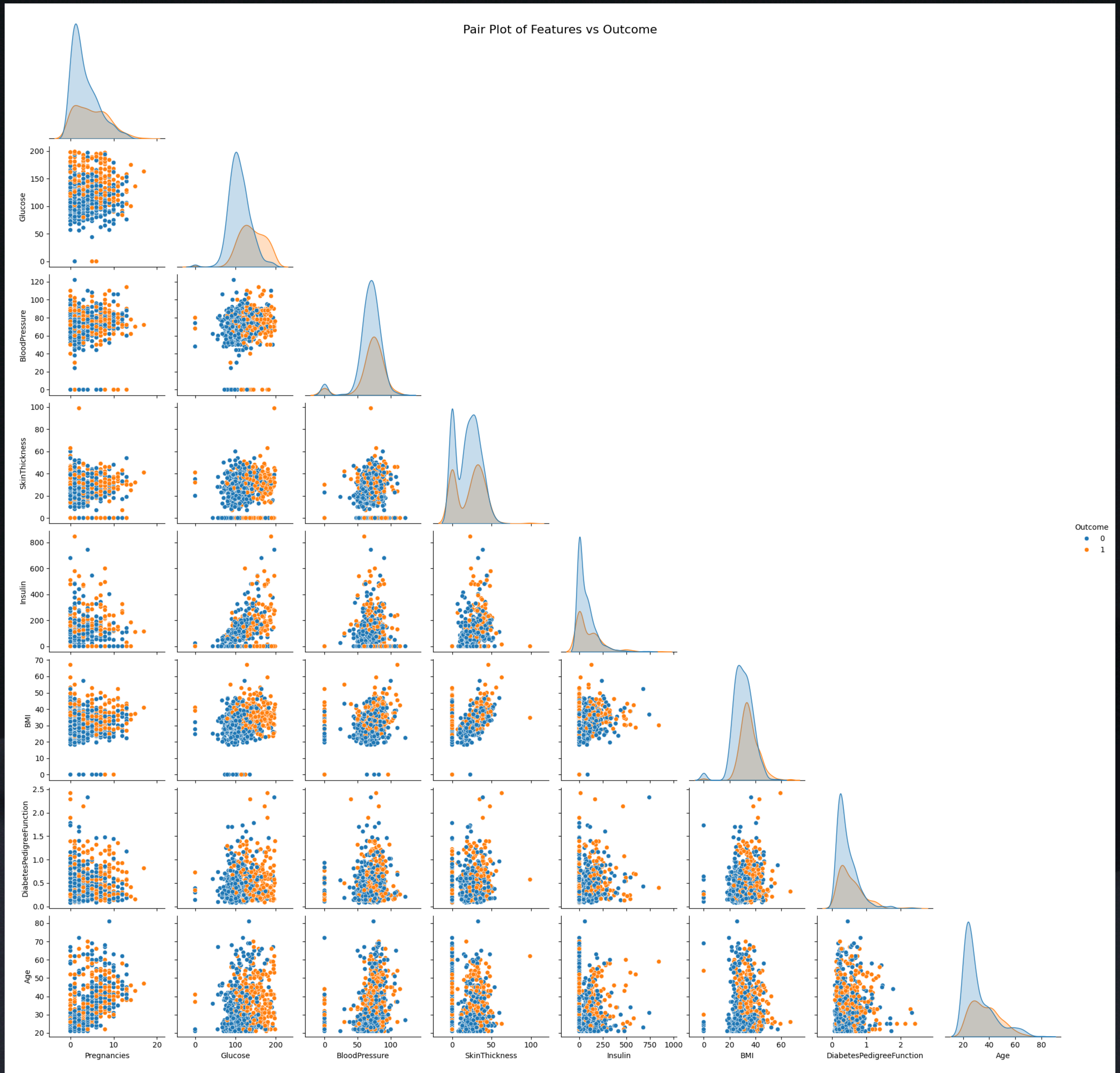
Preprocessing

- *Handel Discretization*
  - Equal Width Binning
  - **Quantile Binning ✓**
  - **K-Means Binning ✓**
  - Round Values to nearest integer

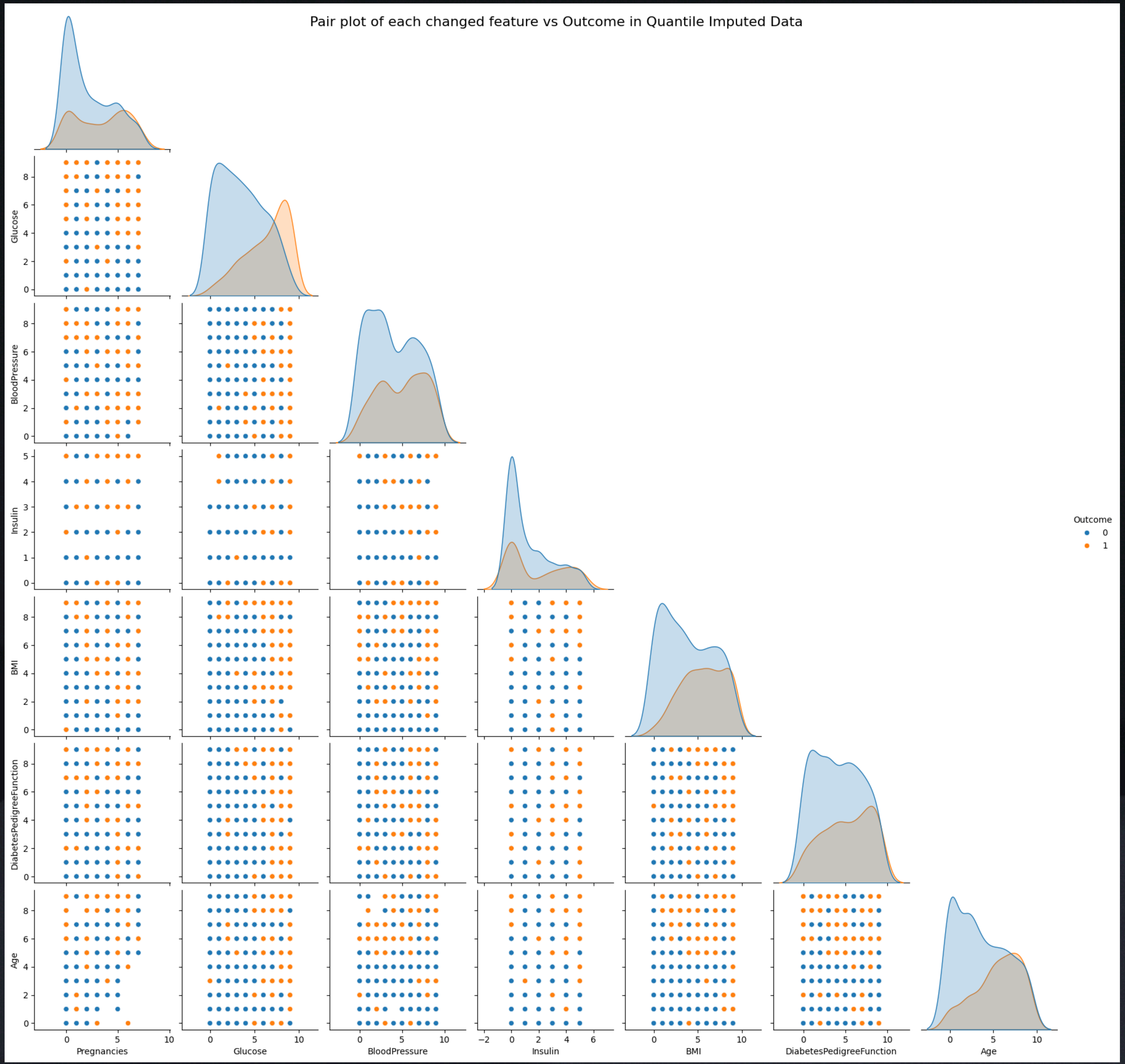
Handle Discretization



Pair Plot



Pair Plot of Features vs Outcome
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Pregnancies
Outcome
0
1

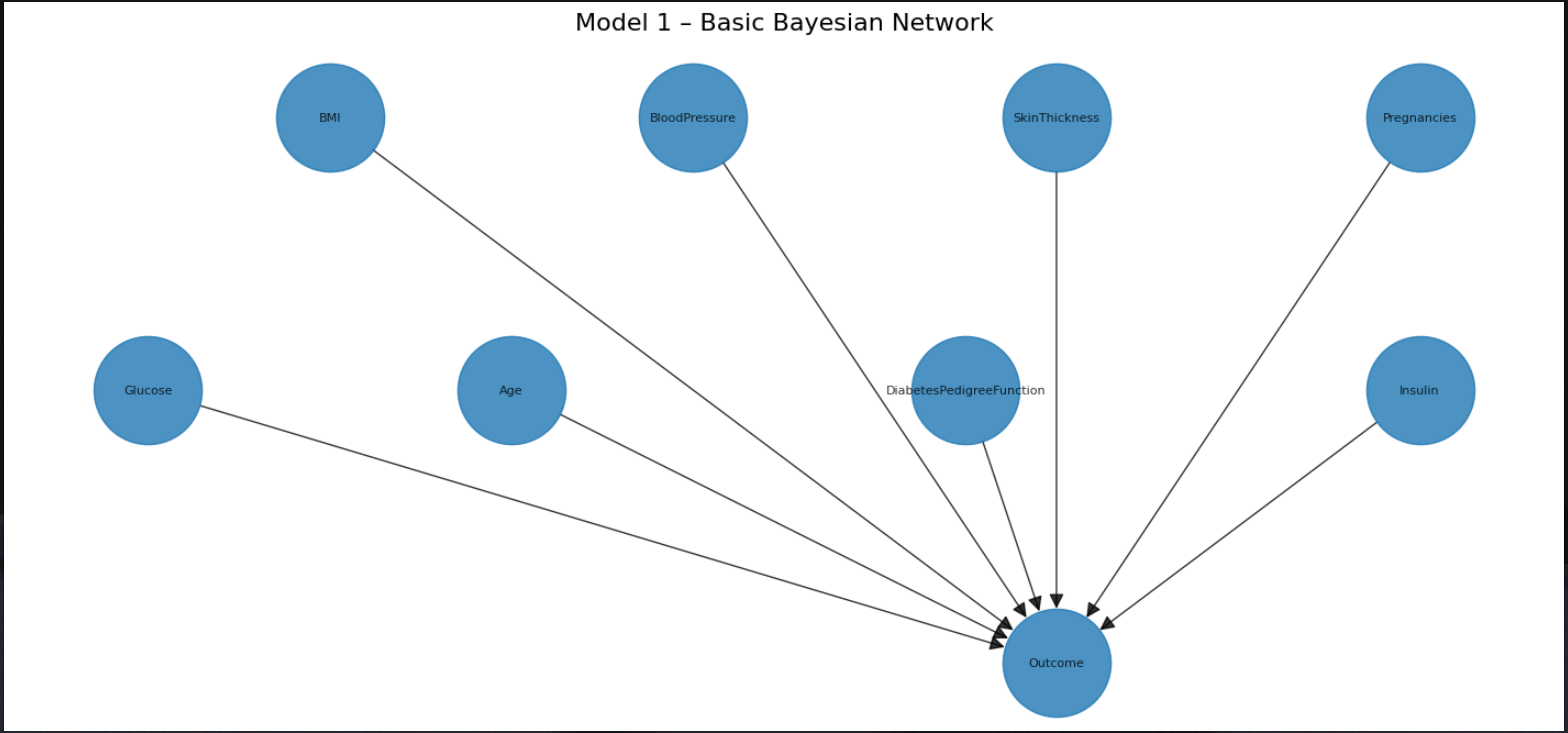


Pair plot of each changed feature vs Outcome in Quantile Imputed Data
Glucose
BloodPressure
Insulin
BMI
DiabetesPedigreeFunction
Age
Pregnancies
Outcome
0
1

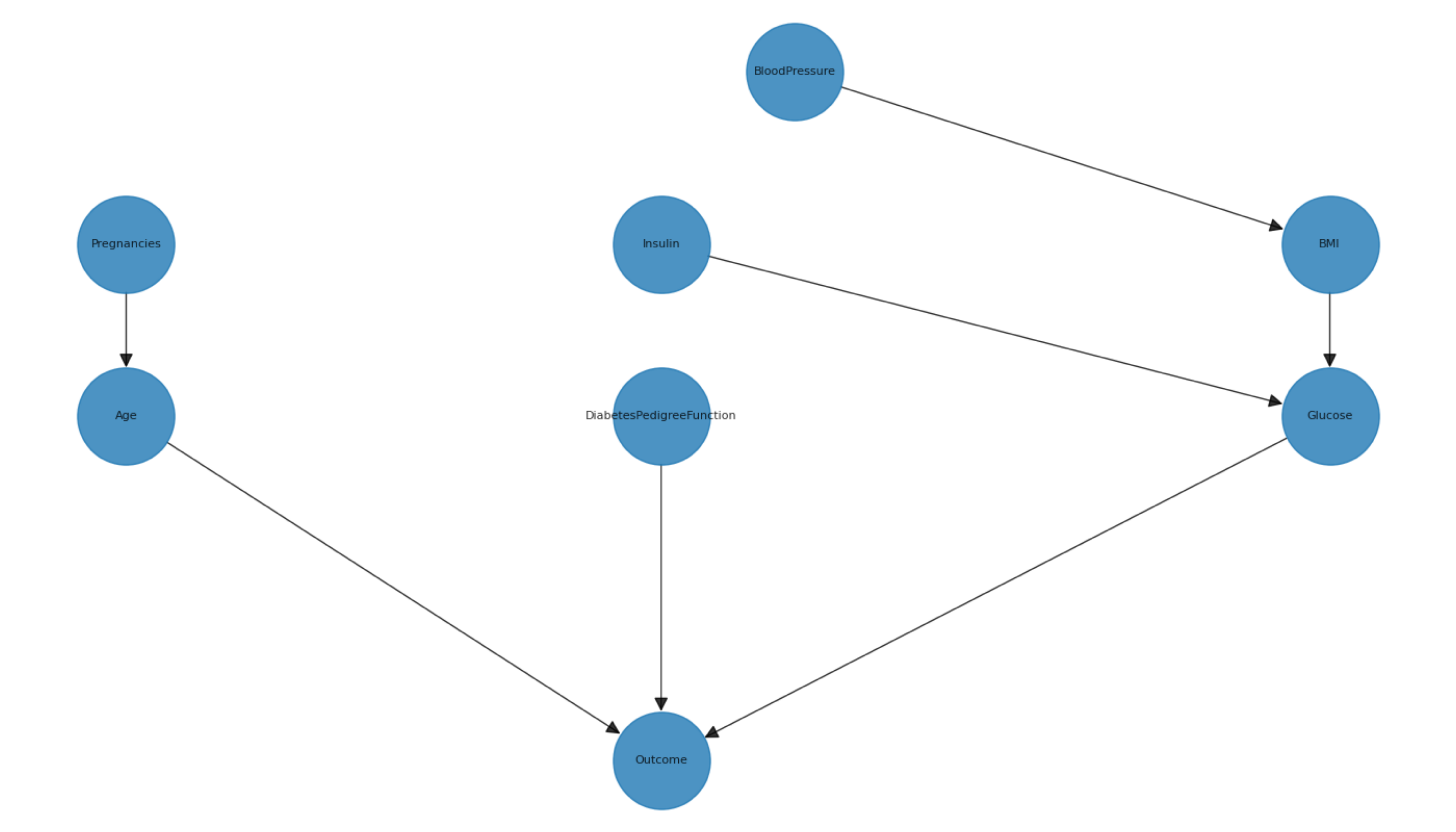
Training Bayesian Network

1. *Define the structure of the Bayesian Network which there is three options here:*
    - we can define the structure manually based on domain knowledge.Quantile Binning
    - we can use structure learning algorithms to learn the structure from data.
    - we can use a hybrid approach where we define some parts of the
2. *Find the Conditional Probability Distributions (CPDs) for each node in the network.*
3. *Perform inference on the Bayesian Network to make analyses with variable elimination. We will perform infrences to analyses the result of different scenarios.*

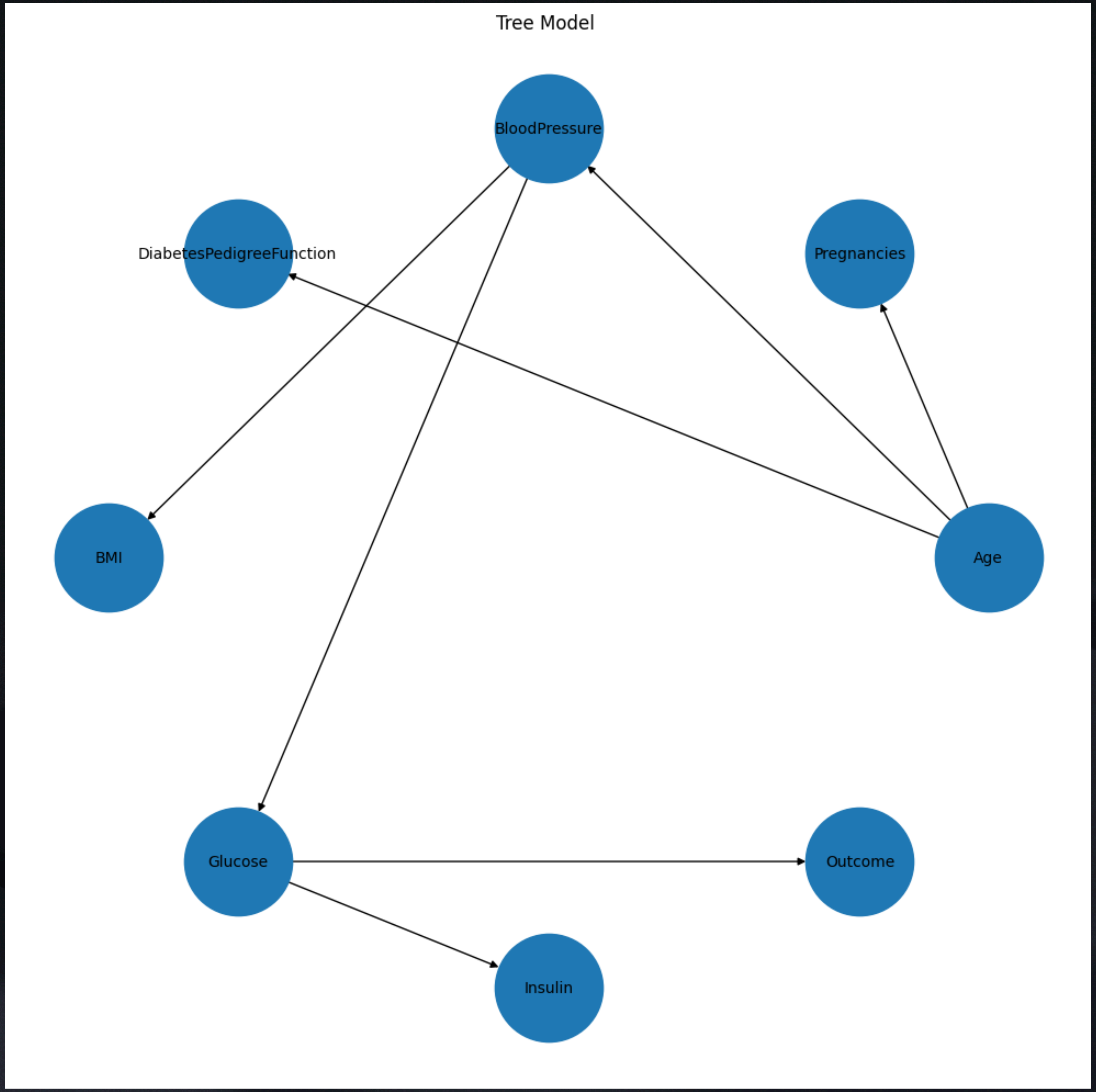
Basic Bayesian Network



Grouped Feature Model

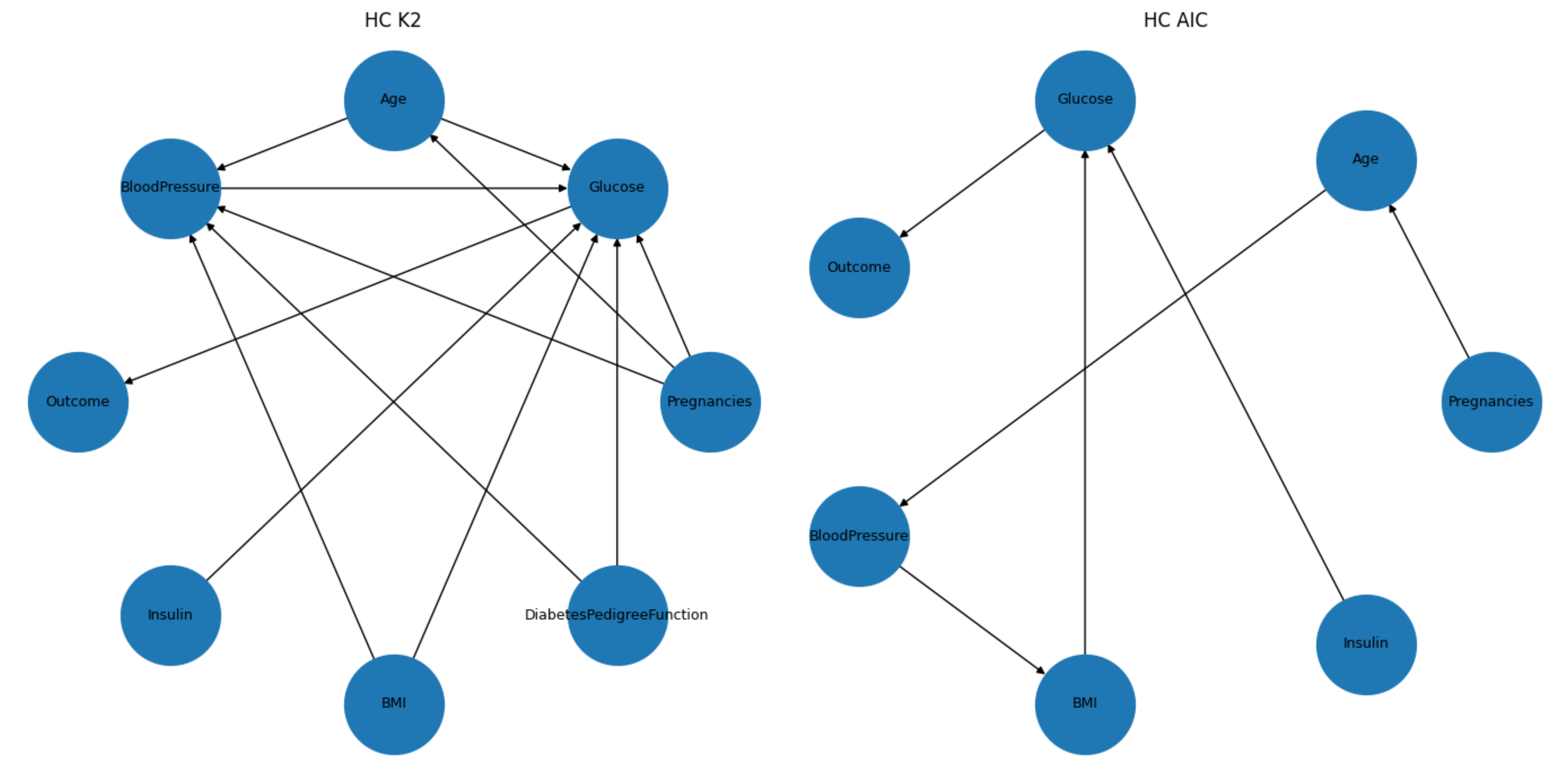


Tree Models



Tree Model
BloodPressure
DiabetesPedigreeFunction
Pregnancies
BMI
Age
Glucose
Outcome
Insulin

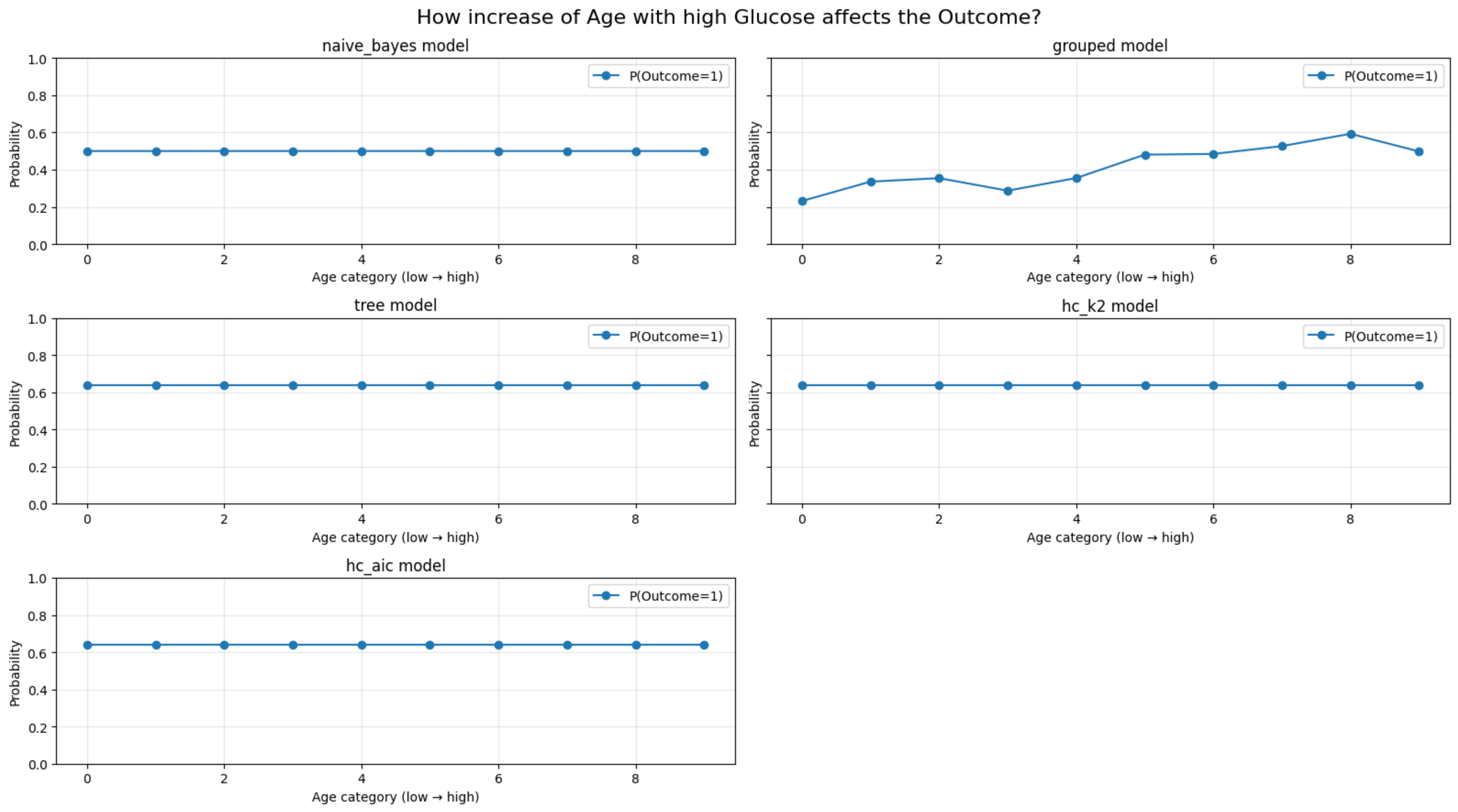
Hill Climbing



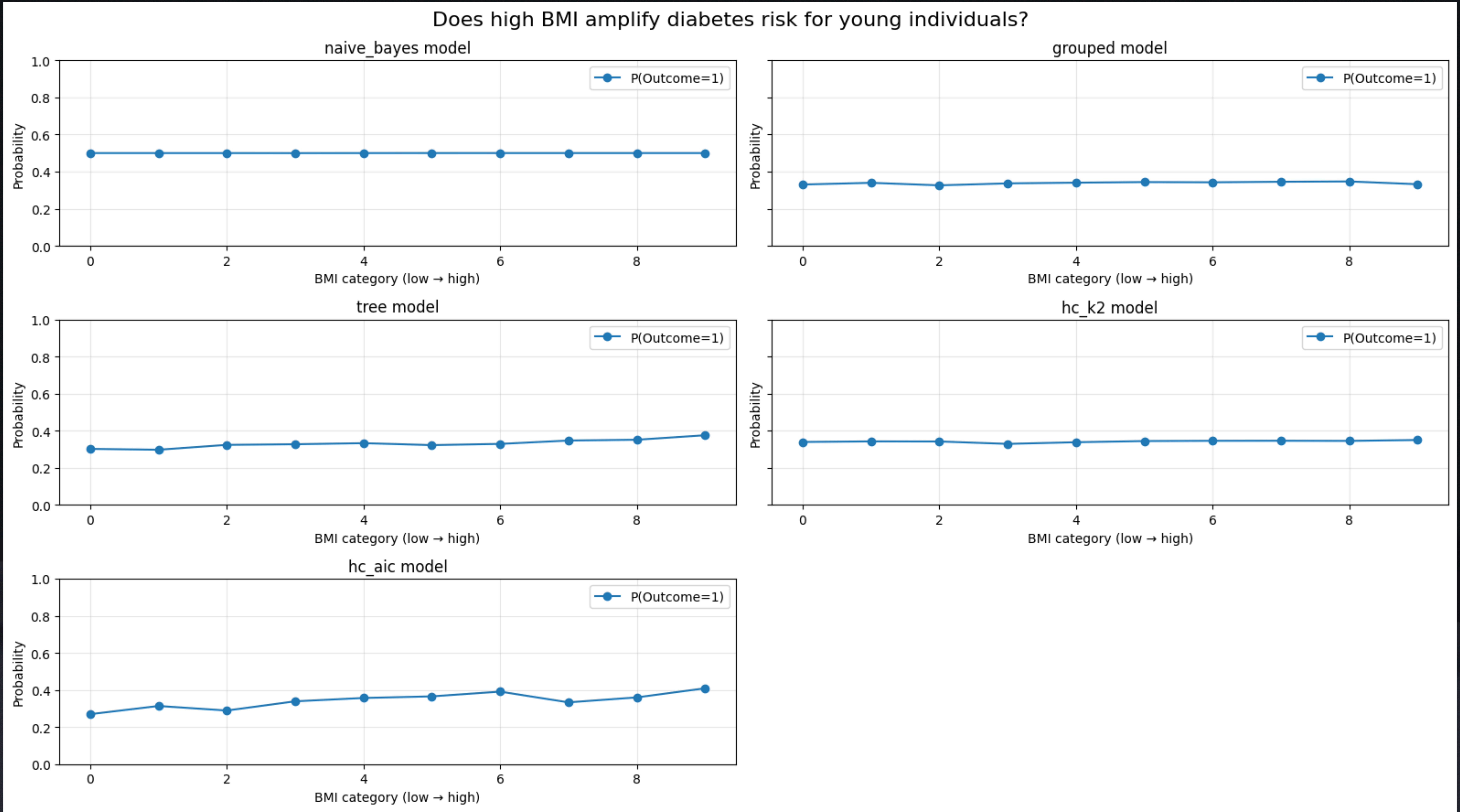
Training

```
Result of  model: hc_k2
Independence assertions num: 16
Local semantics of the current model:
(Pregnancies ⊥ DiabetesPedigreeFunction, Insulin, BMI)
(Age ⊥ DiabetesPedigreeFunction, Insulin, BMI | Pregnancies)
(BloodPressure ⊥ Insulin | DiabetesPedigreeFunction, Age, Pregnancies, BMI)
(Outcome ⊥ DiabetesPedigreeFunction, Pregnancies, BMI, BloodPressure, Insulin, Age | Glucose)
(Insulin ⊥ DiabetesPedigreeFunction, Pregnancies, BMI, BloodPressure, Age)
(BMI ⊥ DiabetesPedigreeFunction, Insulin, Age, Pregnancies)
(DiabetesPedigreeFunction ⊥ Insulin, Age, Pregnancies, BMI)
Checking Markov blankets
The Markov blanket of node Pregnancies is ['DiabetesPedigreeFunction', 'Glucose', 'Insulin', 'Age', 'BMI', 'BloodP
ressure']
The Markov blanket of node Glucose is ['DiabetesPedigreeFunction', 'Insulin', 'Outcome', 'Age', 'Pregnancies', 'BM
I', 'BloodPressure']
The Markov blanket of node Age is ['DiabetesPedigreeFunction', 'Glucose', 'Insulin', 'Pregnancies', 'BMI', 'BloodP
ressure']
The Markov blanket of node BloodPressure is ['DiabetesPedigreeFunction', 'Glucose', 'Insulin', 'Age', 'Pregnancie
s', 'BMI']
The Markov blanket of node Outcome is ['Glucose']
The Markov blanket of node Insulin is ['DiabetesPedigreeFunction', 'Glucose', 'Age', 'Pregnancies', 'BMI', 'BloodP
ressure']
The Markov blanket of node BMI is ['DiabetesPedigreeFunction', 'Glucose', 'Insulin', 'Age', 'Pregnancies', 'BloodP
ressure']
The Markov blanket of node DiabetesPedigreeFunction is ['Glucose', 'Insulin', 'Age', 'Pregnancies', 'BMI', 'BloodP
ressure']
```

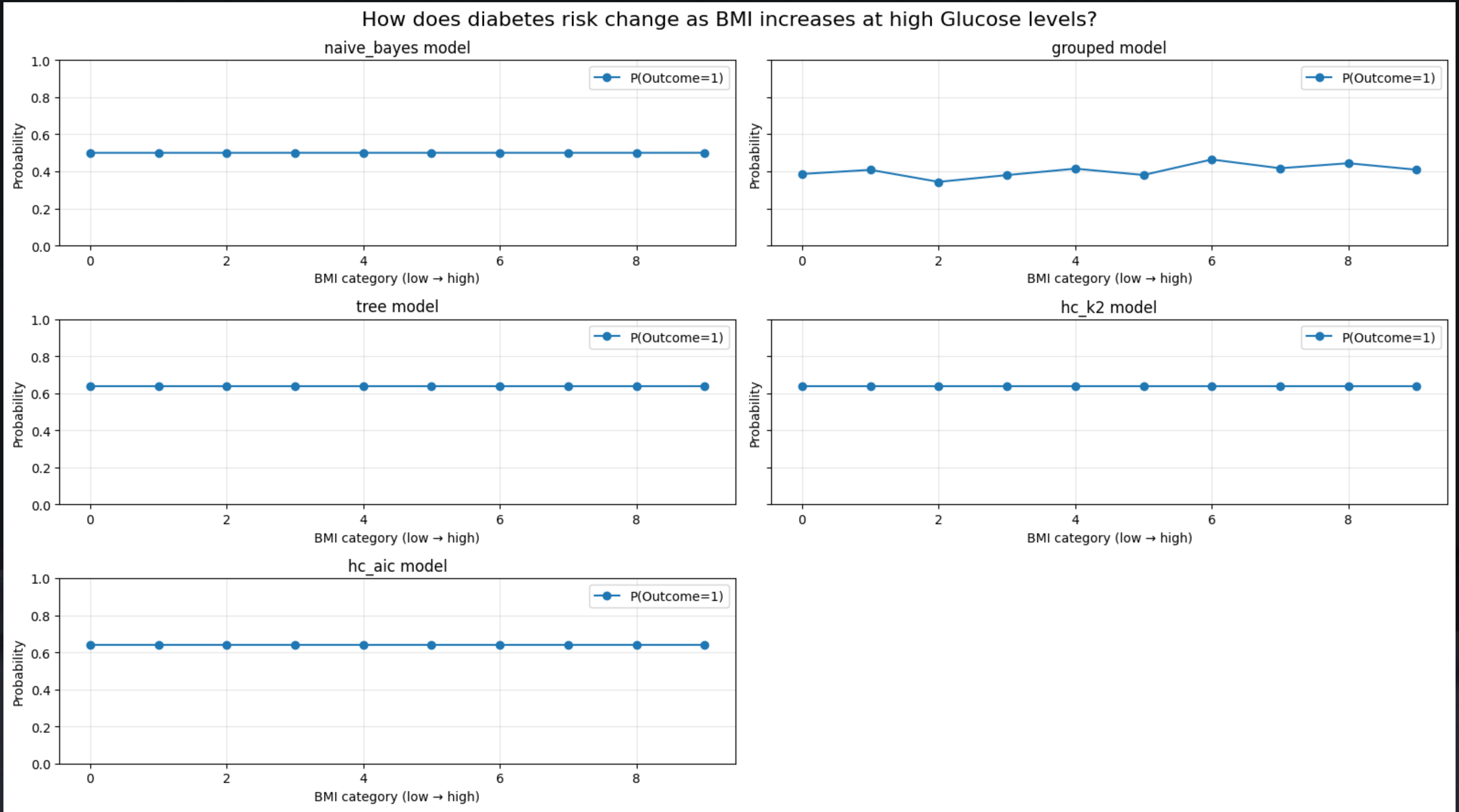
Question 1: How does diabetes risk change when Age increases while Glucose remains high?



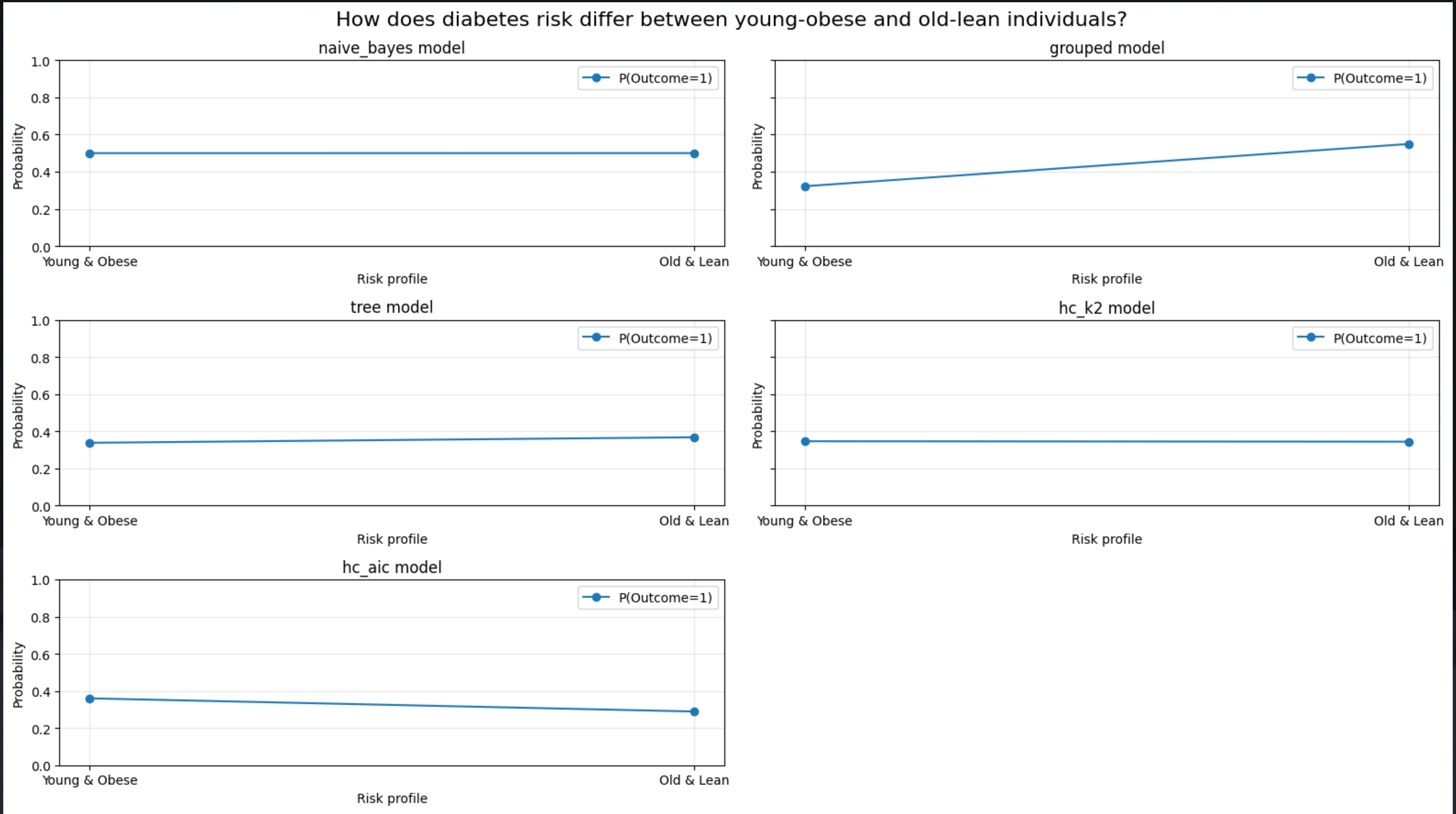
Question 2: Does increasing BMI amplify diabetes risk for young individuals?



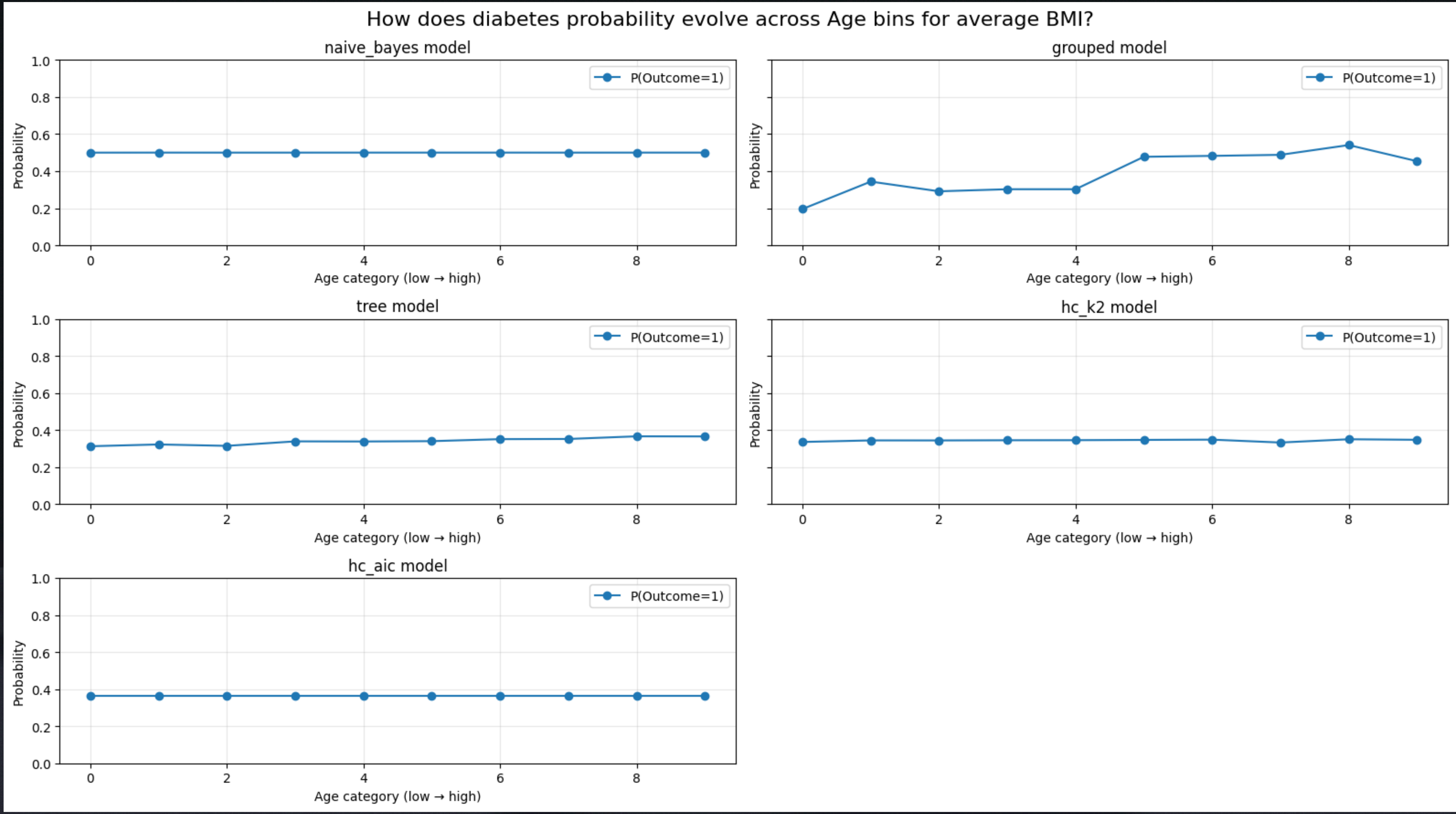
Question 3: How does diabetes risk change as BMI increases at high Glucose levels?



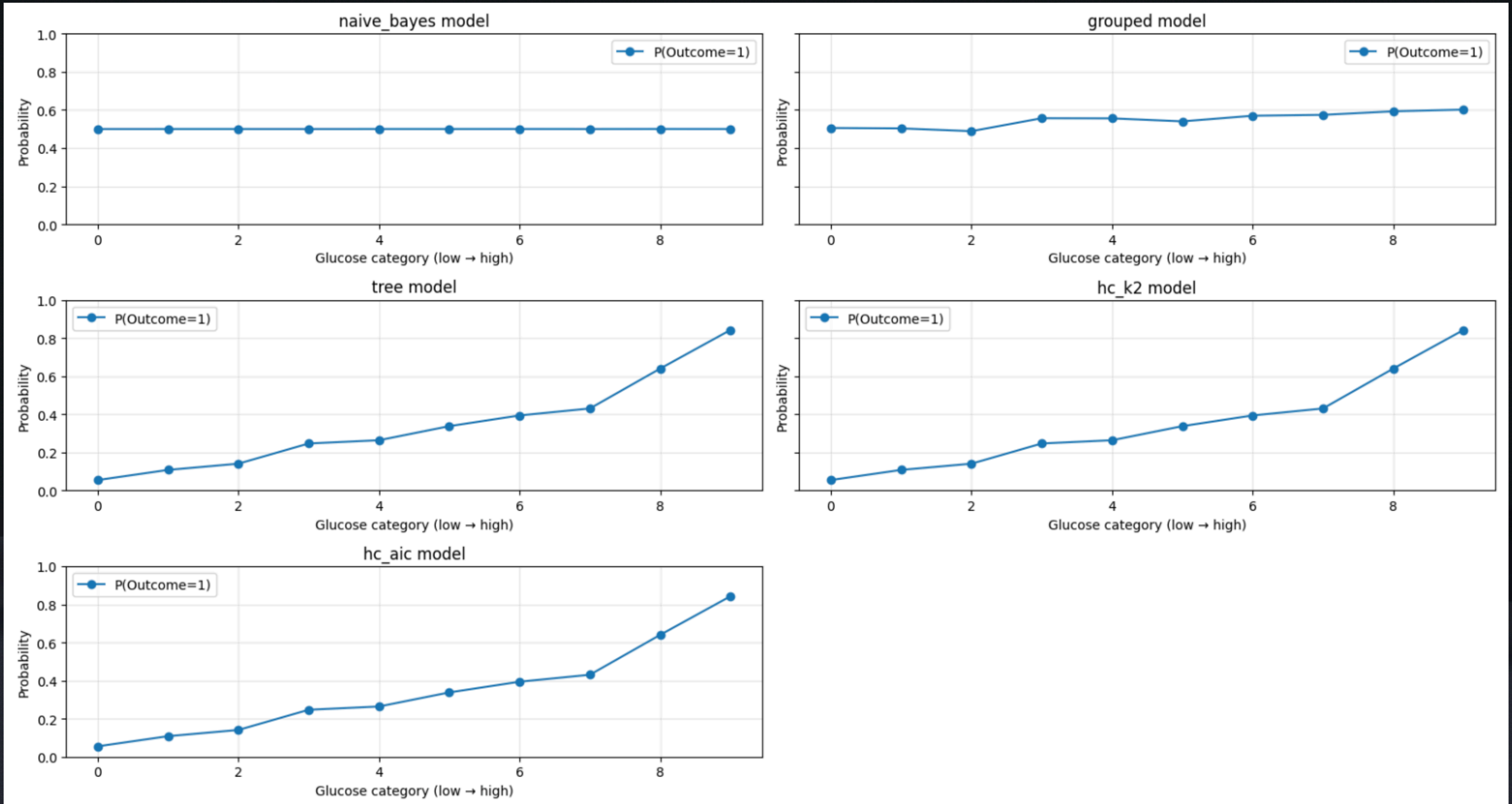
Question 4: How does diabetes risk differ between young-obese and old-lean individuals?



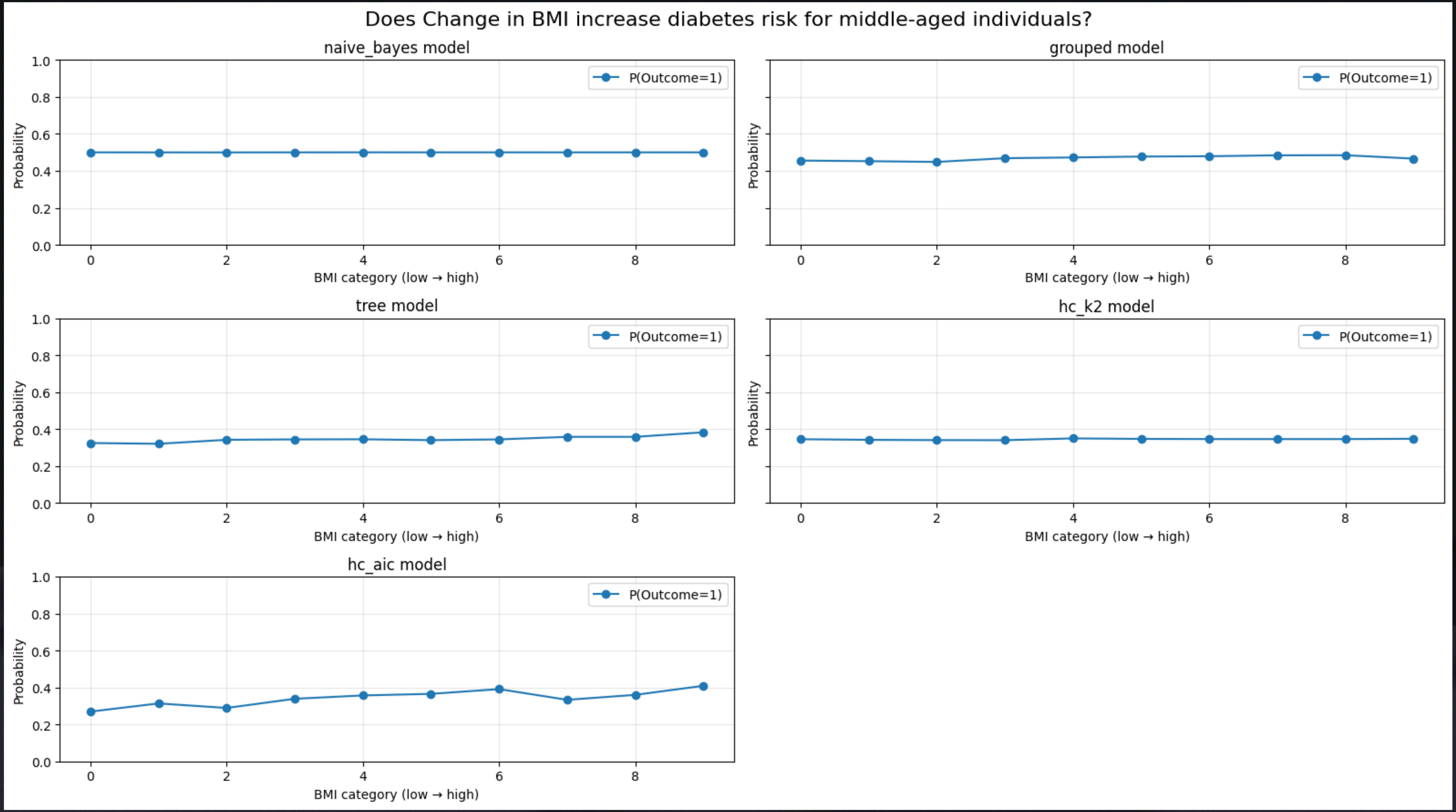
Question 5: How does diabetes probability evolve across Age bins for average BMI?



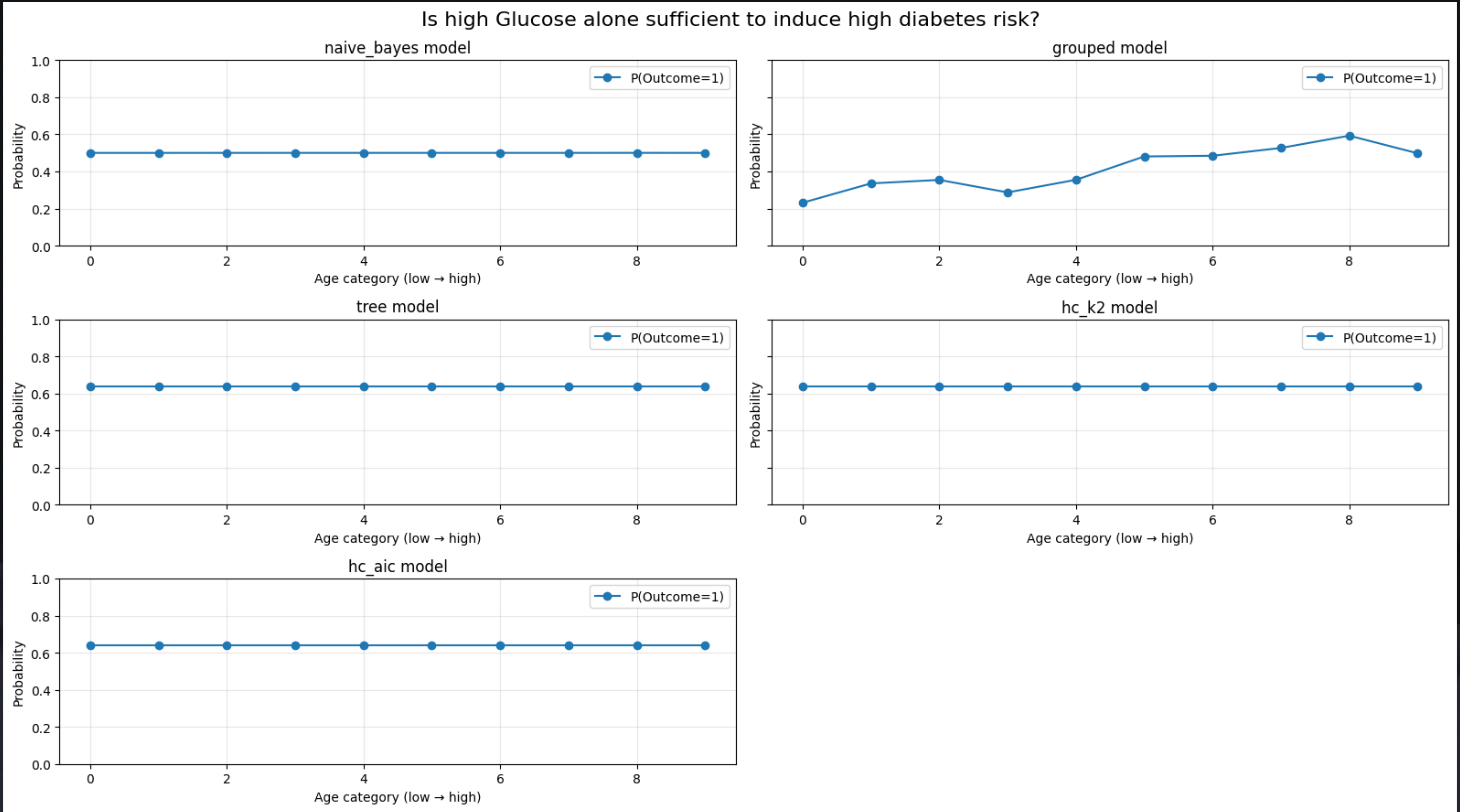
Question 6: At what Glucose level does diabetes risk exceed 50% for elderly individuals?



Question 7: Does Change in BMI increase diabetes risk for middle-aged individuals?

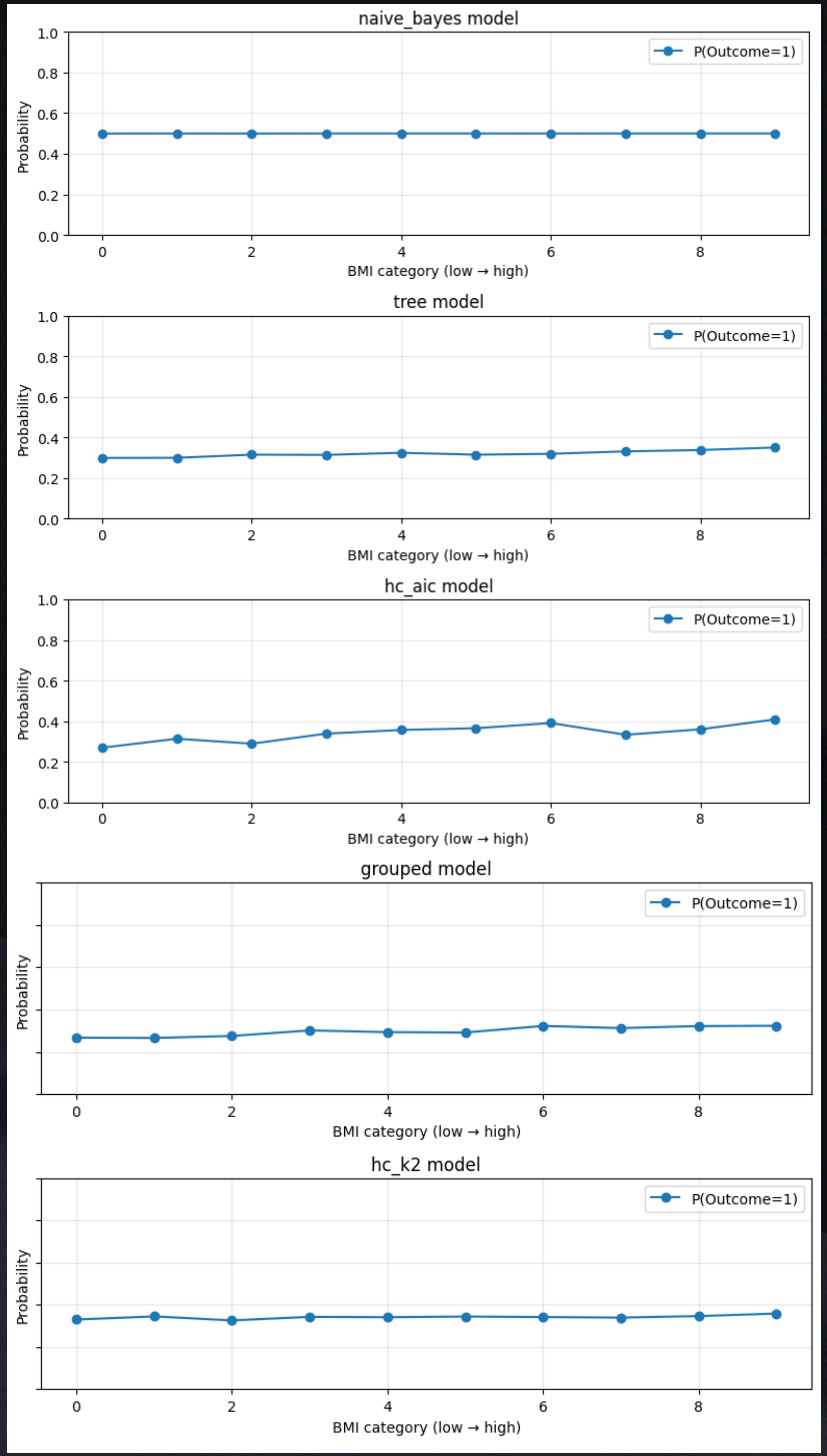


Question 8: Is high Glucose alone sufficient to induce high diabetes risk?

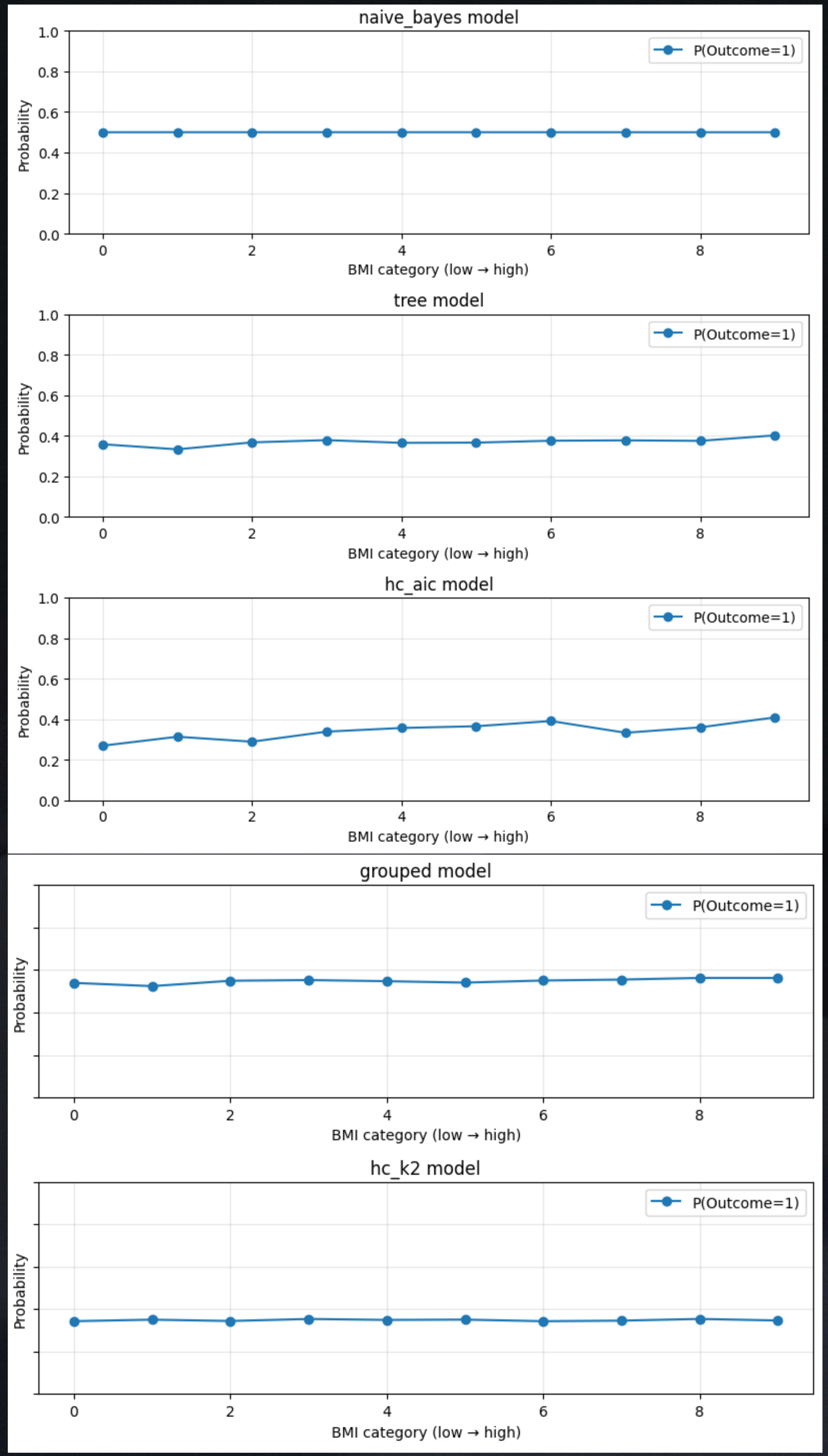


Question 9: Which age group shows the highest sensitivity to BMI changes?

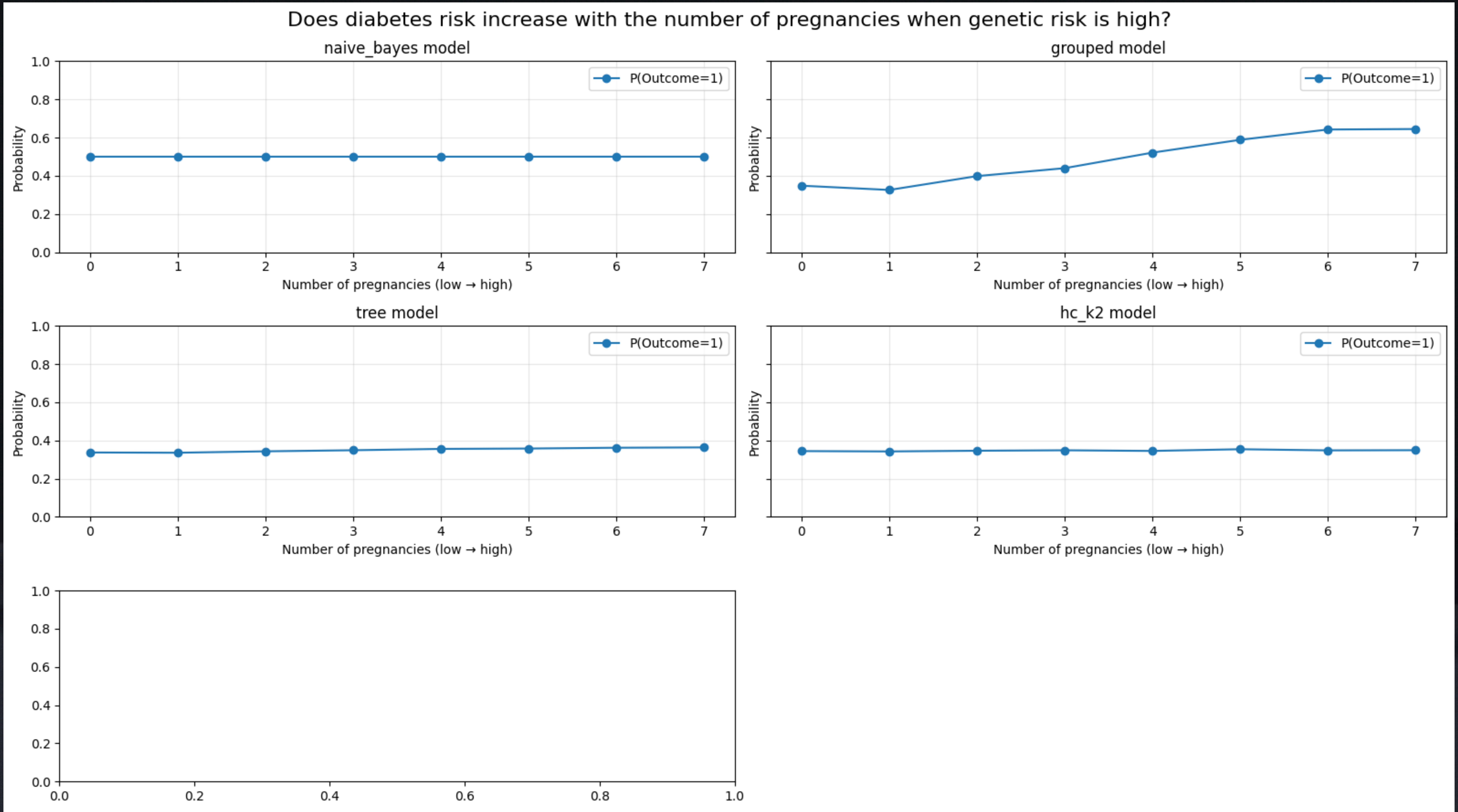
BMI sensitivity for young individuals



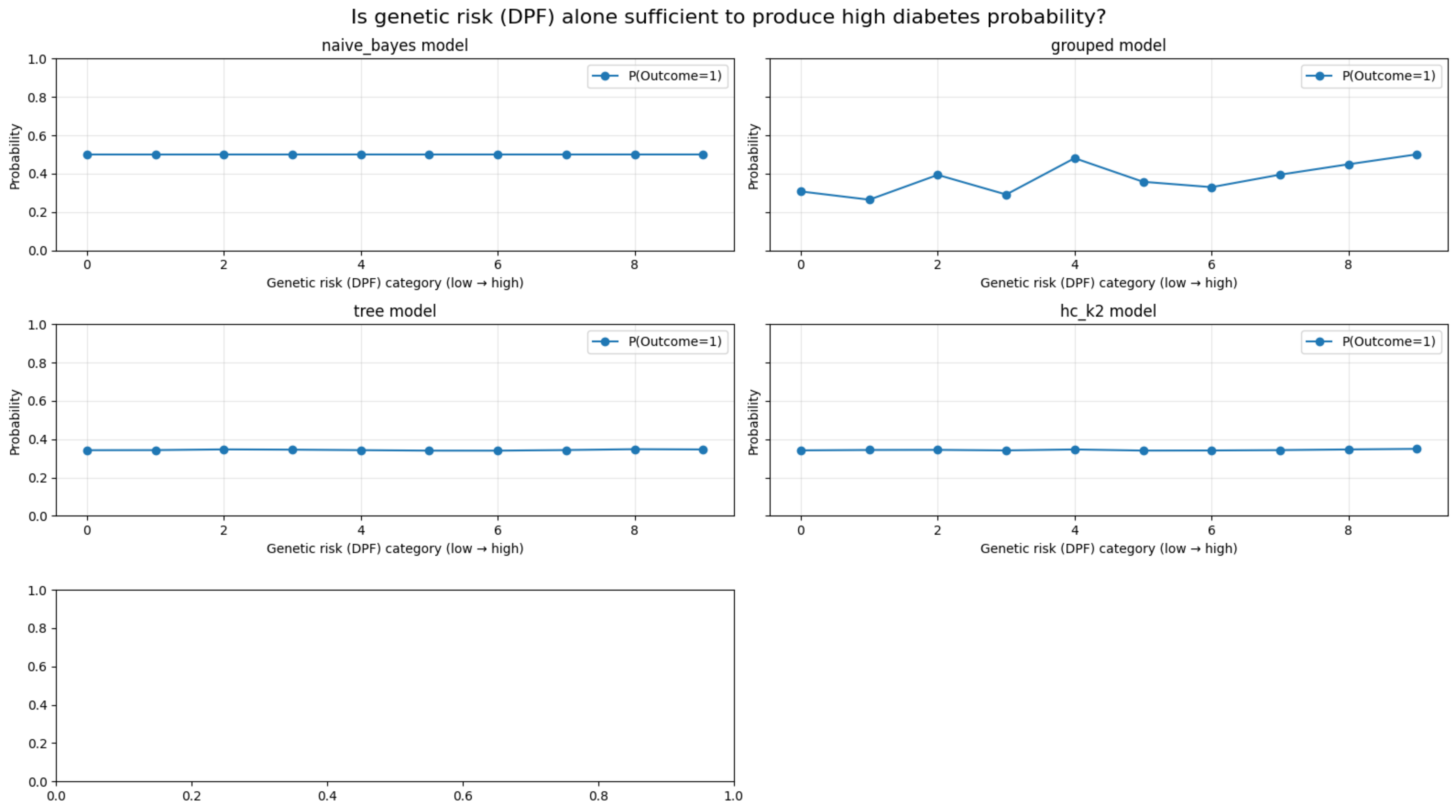
BMI sensitivity for old individuals



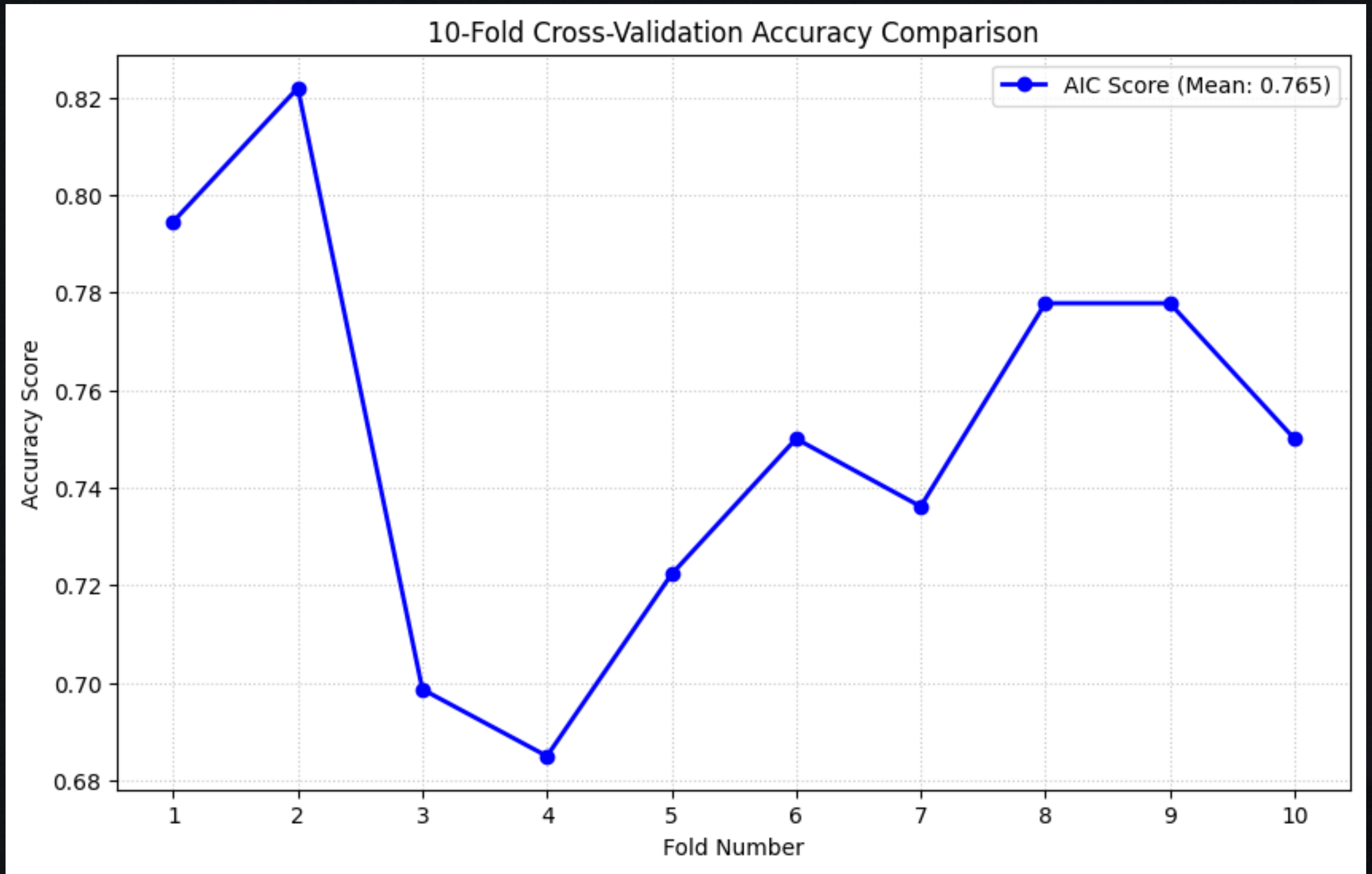
Question 10 : Does diabetes risk increase with the number of pregnancies when genetic risk is high?

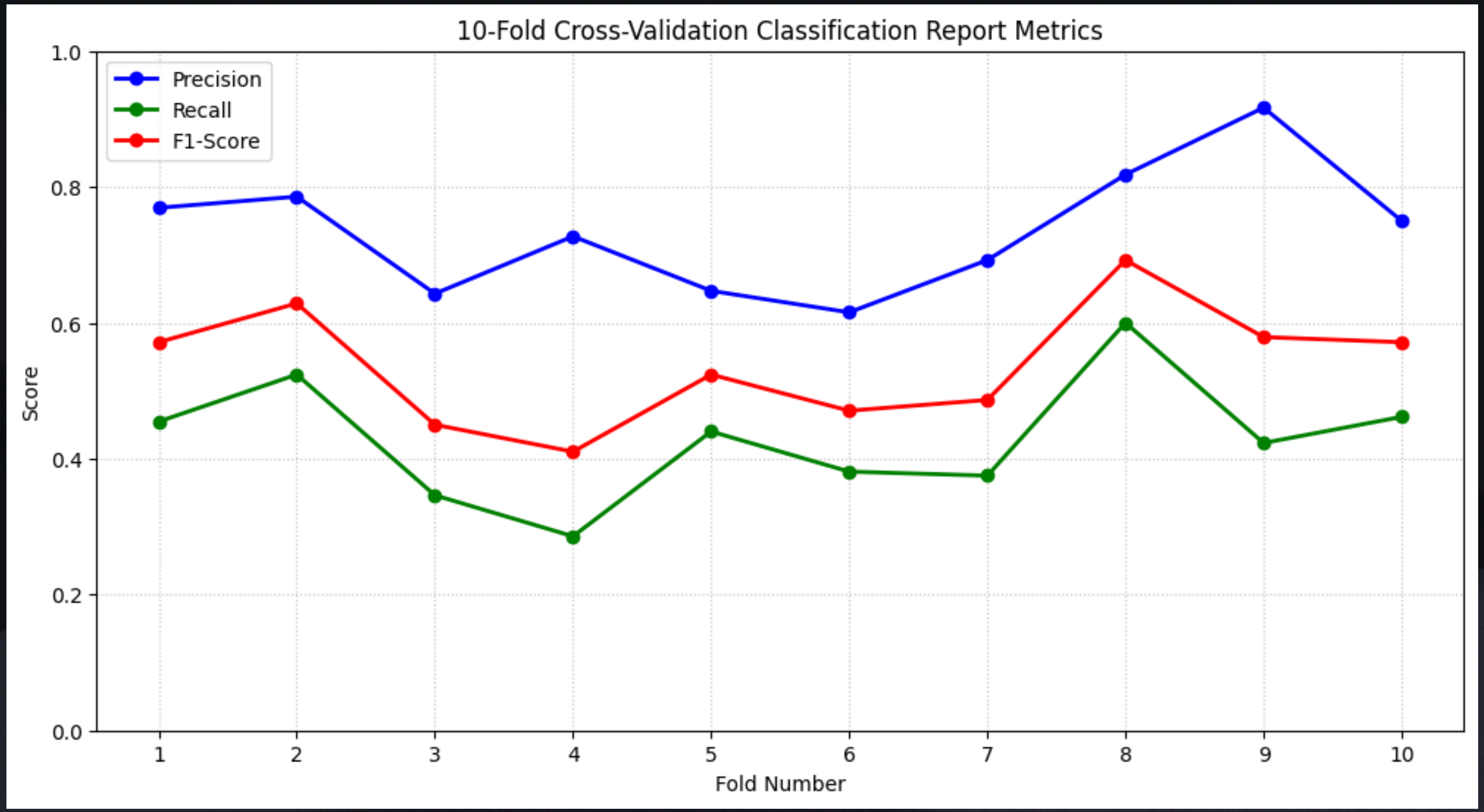


Question 11: Is genetic risk (DPF) alone sufficient to produce high diabetes probability?



Bayesian Network Classification using Hill Climbing





Comparative Performance Table

Model	Sensitivity	Structural Integrity and Clinical Utility	Verdict
Naive Bayes	Very Low	Low	**Failure**: Cannot propagate evidence.
Grouped	Low	Medium	**Insufficient**: Too rigid to capture data nuances.
Tree	High	Medium	**Informative**: Useful but potentially over-simplistic.
HC (K2)	Very High	High	**Aggressive**: Powerful but prone to overconfidence.
HC (AIC)	High	High	**Optimal**: Best balance of complexity and accuracy.

Thank You